

ON THE HYPERBOLIC PRIME NUMBER THEOREM

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ABSTRACT. Friedlander and Iwaniec proved that the number of points of the orbit $\{\gamma i: \gamma \in \mathrm{SL}_2(\mathbb{Z})\}$ lying at a distance $p - 2$ from the origin i of the upper half-plane, with $p \leq x$ prime, is of order $x/\log x$; the upper bound is unconditional, while the lower one rests on a strong hypothesis concerning the distribution of primes in arithmetic progressions. We consider instead square-free distances and prove unconditionally an asymptotic formula, whose main term is of order x . Employing the weighted linear sieve, we also show unconditionally that the square-free distances n with at most 7 prime factors contribute $\gg x/\log x$. Finally, assuming that the sequence $r(n-2)r(n+2)$ has level of distribution x^θ in arithmetic progressions, where $r(n)$ is the number of ways to write n as a sum of two squares, we obtain the same lower bound for the distances with at most N prime factors (the value 7 corresponds to $\theta = 1/6$).

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1. INTRODUCTION

The prime number theorem states that as $x \rightarrow \infty$ we have

$$\pi(x) = \sum_{p \leq x} 1 \sim \int_2^x \frac{dt}{\log t}.$$

In this paper we consider an analogous problem for the hyperbolic plane. Let $\mathbb{H} = \{z = x + iy : x \in \mathbb{R}, y \in \mathbb{R}^+\}$ be the upper half-plane, acted on by the modular group

$$\Gamma = \mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} : x_1, x_2, x_3, x_4 \in \mathbb{Z}, x_1x_4 - x_2x_3 = 1 \right\}$$

by fractional linear transformations. As a distance function on $\mathbb{H}$ we take

$$u(z, w) = \frac{|z - w|^2}{\mathrm{Im} z \mathrm{Im} w},$$

and we choose $z = i$ as the origin, the orbit $\{\gamma i : \gamma \in \Gamma\}$ plays the role of the integers. Similarly to [2, (1.8)] a direct computation gives

$$u(\gamma i, i) + 2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 \quad \text{for } \gamma = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} \in \Gamma.$$

Following [2], for a positive integer n we say that γi is at distance $n - 2$ from the origin if $x_1^2 + x_2^2 + x_3^2 + x_4^2 = n$.

In [2] Friedlander and Iwaniec studied the number $\pi_\Gamma(x)$ of orbit points at a prime distance $p - 2$, $p \leq x$. They showed that $\pi_\Gamma(x) \asymp x/\log x$ (here $f \asymp g$ means that $c_1g \leq f \leq c_2g$ for some positive absolute constants c_1, c_2) under the following hypothesis on primes in arithmetic progressions, which interpolates between the Bombieri–Vinogradov theorem and the Elliott–Halberstam conjecture.

Conjecture 1 (Assumption $A(\theta)$). *Let $E(x, Q)$ denote the remainder term of level Q in Bombieri–Vinogradov theorem that is*

$$\begin{aligned} E(x, Q) &= \sum_{q \leq Q} \max_{(a, q)=1} \max_{y \leq x} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \\ &= \sum_{q \leq Q} \max_{(a, q)=1} \max_{y \leq x} |E(y; q, a)|, \end{aligned}$$

where $\varphi(q)$ is the Euler's totient function, $\psi(x; q, a)$ is the Chebyshev function given by

$$\psi(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n)$$

and $\Lambda(n)$ is the von Mangoldt function. Let also $0 < \theta \leq 1$. Then the bound

$$(A(\theta)) \quad E(x, Q) \ll \frac{x}{(\log x)^A}$$

holds for $Q = x^{\theta-\varepsilon}$, for every $\varepsilon > 0$ and $A > 0$ with an implied constant depending only on ε and A .

The case $\theta = 1/2$ is the Bombieri–Vinogradov theorem and the case $\theta = 1$ is the Elliott–Halberstam conjecture. By [2] we have $\pi_\Gamma(x) \asymp x/\log x$ provided that $A(\theta)$ holds for some $\theta < 1$ sufficiently close to 1 (the upper bound $\pi_\Gamma(x) \ll x/\log x$ being unconditional, hence $A(\theta)$ is essential only for the lower bound). A more general problem, with a polynomial in place of the quadratic form, was considered by Bourgain, Gamburd and Sarnak in [1].

In this paper we replace primes by square-free numbers. Let $s_\Gamma(x)$ be the number of points γi at a distance $n-2$ from the origin with $n \leq x$ and n being free of squares, i.e. $s_\Gamma(x)$ is the number of quadruples $(x_1, x_2, x_3, x_4) \in \mathbb{Z}^4$ such that $x_1^2 + x_2^2 + x_3^2 + x_4^2 = n \leq x$, $x_1 x_4 - x_2 x_3 = 1$ and $\mu^2(n) = 1$. Here μ is the Möbius function, so that $\mu^2(n) = 1$ means that n is square-free. Throughout the paper, χ_4 denotes the non-principal character modulo 4 and $r(m)$ denotes the number of representations of m as a sum of two squares of integers. Our main unconditional result is the following.

Theorem 1. *For $x \geq 2$ and any $\varepsilon > 0$ we have*

$$s_\Gamma(x) = 6Cx + O(x^{11/12+\varepsilon}),$$

where

$$C = \prod_{p>2} \left(1 - \frac{p - \chi_4(p)}{p^2(p + \chi_4(p))} \right), \quad 0 < C < 1.$$

We also consider distances with few prime factors. For an integer $k \geq 1$ let $s_\Gamma^{(k)}(x)$ count the number of (x_1, x_2, x_3, x_4) subject to the conditions of $s_\Gamma(x)$ such that $\omega(n) \leq k$, where $\omega(n)$ is the number of distinct prime factors of n .

Theorem 2. *For $x > 3$ we have*

$$s_\Gamma^{(7)}(x) \gg \frac{x}{\log x}.$$

The number 7 comes from the level of distribution $x^{1/6-\varepsilon}$ of the sequence $r(n-2)r(n+2)$ in arithmetic progressions which is available unconditionally in Lemma 2. We formulate the corresponding hypothesis with a general exponent.

Conjecture 2 (Hypothesis $R(\theta)$). *Let $0 < \theta \leq 1$. For every $\varepsilon > 0$ and $A > 0$ we have*

$$\sum_{\substack{d \leq x^{\theta-\varepsilon} \\ d \text{ odd square-free}}} \left| \sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} r(n-2)r(n+2) - 8g(d)x \right| \ll \frac{x}{(\log x)^A},$$

where

$$g(d) = \frac{1}{d} \prod_{p|d} \frac{p - \chi_4(p)}{p + \chi_4(p)},$$

see also Lemma 2, and an implied constant depending only on ε and A .

By Lemma 2, $R(\theta)$ holds for $\theta = 1/6$. In analogy with the Elliott–Halberstam conjecture one may expect $R(\theta)$ to hold for every $\theta < 1$. Note that $R(\theta)$ concerns the weights $r(n-2)r(n+2)$ only and does not involve primes. Our conditional result is the following.

Theorem 3. *Let $0 < \theta \leq 1$ and assume $R(\theta)$. Then, with $N = \lfloor 2/\theta \rfloor$, for $x > 3$ we have*

$$s_{\Gamma}^{(N)}(x) \gg \frac{x}{\log x}.$$

In particular, $R(1/2)$ implies $s_{\Gamma}^{(4)}(x) \gg x/\log x$.

Theorem 3 is obtained from the linear sieve, see Lemma 4, applied to the sequence $r(n-2)r(n+2)$, whose level of distribution, by Lemma 2 below, is $x^{1/6-\varepsilon}$. Taking $\theta = 1/6$ in the above theorem gives $\omega(n) \leq 12$ unconditionally. Theorem 2 is obtained by adding to this argument the logarithmic weights of Richert [7], which brings the number 12 down to 7.

In a recent paper [8], Shparlinski and Xiao obtain, by rather different methods, an asymptotic formula for the number $S_{\text{sq}}(T)$ of matrices

$$\gamma = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} \in \text{SL}_2(\mathbb{Z}), \quad \max_{1 \leq i \leq 4} |x_i| \leq T,$$

for which $n = x_1^2 + x_2^2 + x_3^2 + x_4^2$ is square-free. Their treatment is considerably more general than ours and proceeds by a δ -method combined with the weighted linear sieve.

The two problems differ in the way the matrices are ordered: in [8] they are ordered by the sup-norm of the entries, whereas we order them by the value of the quadratic form, that is via $n \leq x$. This ordering connects the problem to the sequence $r(n-2)r(n+2)$, so that we may follow the approach of [2], allowing us to get a better level of distribution and, therefore, fewer prime factors via Lemma 2.

2. AUXILIARY STATEMENTS

2.1. Representations as a sum of two squares. Let us first recall some known results about $r(n)$. In what follows χ_4 is the non-principal Dirichlet character modulo 4 that is completely multiplicative function given by

$$\chi_4(p) = \begin{cases} 1, & p \equiv 1 \pmod{4}, \\ -1, & p \equiv 3 \pmod{4}, \\ 0, & p = 2. \end{cases}$$

We use the classical formula

$$r(m) = 4 \sum_{d|m} \chi_4(d), \quad m \geq 1,$$

which in particular gives the trivial bound $r(m) \leq 4\tau(m) \ll m^\varepsilon$, where τ is the divisor function.

The following two results are Lemma 4.1 and Lemma 4.2 of [2], respectively, for references see also results of Smith [9], [10].

Lemma 1. *Let $(a, q) = 1$. Then for $q \not\equiv 0 \pmod{4}$ we have*

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} r(n) = \frac{\pi x}{q} \prod_{p|q} \left(1 - \frac{\chi_4(p)}{p}\right) + O(q^{-1/2} x^{2/3+\varepsilon})$$

and the main term is multiplied by $1 + \chi_4(a)$ for $q \equiv 0 \pmod{4}$.

Lemma 2. *Let d be an odd positive integer and let $\varepsilon > 0$. Then*

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} r(n-2)r(n+2) = 8g(d)x + O(d^{-1/2} x^{11/12+\varepsilon} + x^{1/2+\varepsilon}),$$

where

$$g(d) = \frac{1}{d} \prod_{p|d} \frac{p - \chi_4(p)}{p + \chi_4(p)}$$

and the implied constant depends only on ε . In particular the error term is $O(x^{11/12+\varepsilon})$ uniformly in d .

Lemma 4.2 of [2] is stated with the error term $O(x^{11/12+\varepsilon})$, however we need the refined form of Lemma 2, which the proof in [2] in fact gives. The lemma is stated in [2] for all odd d , not only for square-free d , and we shall apply it with d a perfect square. Below we recall the argument, indicating the changes in the error term in comparison with [2].

Proof. Recall that for $m \equiv 1 \pmod{4}$ we have

$$r(m) = 8 \sum_{\substack{k|m \\ k < \sqrt{m}}} \chi_4(k) + 4\chi_4(\sqrt{m}),$$

where the last term vanishes if m is not a square. Since $n \equiv 3 \pmod{4}$ we have $n - 2 \equiv 1 \pmod{4}$, so

$$r(n - 2) = 8 \sum_{\substack{k|n-2 \\ k < \sqrt{n-2}}} \chi_4(k) + 4\chi_4(\sqrt{n-2}),$$

the last term occurring only when $n - 2$ is a square. When summed over $n \leq x$ the last term contributes

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} r(n+2)\chi_4(\sqrt{n-2}) \ll \sum_{m \leq x} m^\varepsilon \chi_4(\sqrt{m}) \ll x^{1/2+\varepsilon},$$

which is negligible for our goals.

Writing $k \mid n - 2$ as $n \equiv 2 \pmod{k}$ we therefore obtain

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} r(n-2)r(n+2) = 8 \sum_{\substack{k < \sqrt{x-2} \\ (k,d)=1}} \chi_4(k) \sum_{\substack{k^2+2 < n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d} \\ n \equiv 2 \pmod{k}}} r(n+2) + O(x^{1/2+\varepsilon}),$$

where only odd k occur, since $\chi_4(k) = 0$ otherwise.

Note that as $n \equiv 0 \pmod{d}$ and $n \equiv 2 \pmod{k}$ we have $(k, d) = 1$. Indeed, assume $(k, d) = \delta > 1$ and write $d = d'\delta$, $k = k'\delta$. From $n \equiv 0 \pmod{d}$ and $n \equiv 2 \pmod{k}$ we get $n = ud'\delta$ and $n = vk'\delta + 2$ for some integers u, v , thus $\delta(ud' - vk') = 2$ and so $\delta \mid 2$. But $\delta \mid d$ and d is odd, therefore $\delta = 1$.

For $(k, d) = 1$ and k odd, $n + 2$ runs over a residue class modulo $4dk$ which is coprime to $4dk$ and congruent to 1 modulo 4, therefore we can apply Lemma 1 and get

$$\begin{aligned} \sum_{\substack{k^2+2 < n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d} \\ n \equiv 2 \pmod{k}}} r(n+2) &= \sum_{\substack{k^2+4 < m \leq x+2 \\ m \equiv a \pmod{4dk}}} r(m) \\ &= \frac{\pi(x - k^2)}{2dk} h(d)h(k) + O((dk)^{-1/2}(x^{2/3+\varepsilon} + k^{4/3+\varepsilon}) + (dk)^{-1+\varepsilon'}), \end{aligned}$$

where we used

$$\prod_{p|4dk} \left(1 - \frac{\chi_4(p)}{p}\right) = h(d)h(k)$$

where $h(m) = \prod_{p|m} (1 - \chi_4(p)/p)$, and we used that $\chi_4(2) = 0$ together with $(k, d) = 1$. Here a denotes the class given by the Chinese remainder theorem, and the main term is multiplied by $1 + \chi_4(a) = 2$, since $4dk \equiv 0 \pmod{4}$ and $a \equiv 1 \pmod{4}$. The term $O((dk)^{-1+\varepsilon'})$ comes from the length of the interval being $x - k^2 - 2$.

As $k \ll \sqrt{x}$ the error term contribution is

$$d^{-1/2} \sum_{k \ll \sqrt{x}} (x^{2/3+\varepsilon} k^{-1/2} + k^{5/6+\varepsilon} + d^{-1/2+\varepsilon'} k^{-1+\varepsilon'}) \ll d^{-1/2} x^{11/12+\varepsilon}.$$

Thus we have achieved so far

$$\begin{aligned} \sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} r(n-2)r(n+2) &= 4\pi \frac{h(d)}{d} \sum_{\substack{k < \sqrt{x} \\ (k,d)=1}} \frac{\chi_4(k)}{k} h(k)(x - k^2) \\ &\quad + O(d^{-1/2} x^{11/12+\varepsilon} + x^{1/2+\varepsilon}). \end{aligned}$$

Completing the sum over k to infinity and using that $h(p) = 1 - \chi_4(p)/p$ we get

$$\begin{aligned} \sum_{(k,d)=1} \frac{\chi_4(k)h(k)}{k} &= \prod_{p \nmid d} \left(1 + \left(1 - \frac{\chi_4(p)}{p} \right) \sum_{j \geq 1} \frac{\chi_4(p)^j}{p^j} \right) \\ &= \prod_{p \nmid d} \left(1 + \frac{\chi_4(p)}{p} \right) = \frac{2}{\pi} \prod_{p \nmid d} \left(1 + \frac{\chi_4(p)}{p} \right)^{-1}, \end{aligned}$$

where we used that

$$\begin{aligned} \prod_p \left(1 + \frac{\chi_4(p)}{p} \right) &= \prod_p \left(1 - \frac{\chi_4(p)}{p} \right)^{-1} \left(1 - \frac{\chi_4^2(p)}{p^2} \right) \\ &= \prod_p \left(1 - \frac{\chi_4(p)}{p} \right)^{-1} \prod_{p > 2} \left(1 - \frac{1}{p^2} \right) \\ &= \frac{L(1, \chi_4)}{\zeta(2)} \left(1 - \frac{1}{4} \right)^{-1} = \frac{2}{\pi}. \end{aligned}$$

Therefore, the contribution of this completed to infinity sum is x times

$$\begin{aligned} \frac{8h(d)}{d} \prod_{p \nmid d} \left(1 + \frac{\chi_4(p)}{p} \right)^{-1} &= \frac{8}{d} \prod_{p \nmid d} \left(1 - \frac{\chi_4(p)}{p} \right) \left(1 + \frac{\chi_4(p)}{p} \right)^{-1} \\ &= \frac{8}{d} \prod_{p \nmid d} \frac{p - \chi_4(p)}{p + \chi_4(p)} = 8g(d), \end{aligned}$$

which is our expected main term. We note that $0 \leq g(p) < 1$ for every odd prime $p > 2$, and that

$$g(p) = \frac{1}{p} - \frac{2\chi_4(p)}{p(p + \chi_4(p))} = \frac{1}{p} + O\left(\frac{1}{p^2}\right), \quad g(p^2) = \frac{p - \chi_4(p)}{p^2(p + \chi_4(p))} \leq \frac{2}{p^2}.$$

It remains to show that the tail and terms coming from k^2 are small and hence go to the error term. Notice that elementarily we have

$$\sum_{\substack{k \leq t \\ (k,d)=1}} \chi_4(k) = \sum_{d' \mid d} \mu(d') \chi_4(d') \sum_{m \leq t/d'} \chi_4(m) \ll \tau(d).$$

For $p \nmid 2d$ and $\text{Res} > 1$ we have

$$\sum_{j \geq 0} \frac{\chi_4(p^j) h(p^j)}{p^{js}} = 1 + \left(1 - \frac{\chi_4(p)}{p}\right) \frac{\chi_4(p) p^{-s}}{1 - \chi_4(p) p^{-s}} = \frac{1 - p^{-s-1}}{1 - \chi_4(p) p^{-s}},$$

so that, comparing coefficients in

$$\sum_{(k,d)=1} \frac{\chi_4(k) h(k)}{k^s} = \prod_{p \nmid d} \left(1 - \frac{\chi_4(p)}{p^s}\right)^{-1} \prod_{p \mid 2d} \left(1 - \frac{1}{p^{s+1}}\right),$$

for every k with $(k, d) = 1$ one has

$$\chi_4(k) h(k) = \sum_{k' \mid k} \chi_4(k/k') \frac{\mu(k')}{k'},$$

where k' runs over odd square-free numbers. Therefore

$$\sum_{\substack{k \leq t \\ (k,d)=1}} \chi_4(k) h(k) = \sum_{\substack{k' \leq t \\ (k', 2d)=1}} \frac{\mu(k')}{k'} \sum_{\substack{u \leq t/k' \\ (u,d)=1}} \chi_4(u)$$

and consequently

$$\sum_{\substack{k \leq t \\ (k,d)=1}} \chi_4(k) h(k) \ll \tau(d) \sum_{k' \leq t} \frac{|\mu(k')|}{k'} \ll \tau(d) \log t.$$

Thus, by partial summation we have

$$\sum_{\substack{k > \sqrt{x} \\ (k,d)=1}} \frac{\chi_4(k)}{k} h(k) \ll \frac{\tau(d) \log x}{\sqrt{x}} + \tau(d) \int_{\sqrt{x}}^{\infty} \frac{\log t}{t^2} dt \ll \frac{\tau(d) \log x}{\sqrt{x}}$$

and the tail contribution can be bounded as

$$4\pi x \frac{h(d)}{d} \sum_{\substack{k > \sqrt{x} \\ (k,d)=1}} \frac{\chi_4(k)}{k} h(k) \ll x^{1/2+\varepsilon}.$$

It remains to bound the contribution of terms involving k^2 , which up to a constant factor time $h(d)/d$ is

$$\begin{aligned} \sum_{\substack{k \leq \sqrt{x} \\ (k,d)=1}} \chi_4(k)h(k)k &= \sqrt{x} \sum_{\substack{k \leq \sqrt{x} \\ (k,d)=1}} \chi_4(k)h(k) - \int_1^{\sqrt{x}} \left(\sum_{\substack{k \leq t \\ (k,d)=1}} \chi_4(k)h(k) \right) dt \\ &\ll \tau(d)\sqrt{x} \log x, \end{aligned}$$

therefore

$$\frac{h(d)}{d} \sum_{\substack{k \leq \sqrt{x} \\ (k,d)=1}} \chi_4(k)h(k)k \ll \frac{h(d)}{d} \tau(d)\sqrt{x} \log x \ll x^{1/2+\varepsilon}.$$

Finally, we may conclude

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} r(n-2)r(n+2) = 8g(d)x + O(d^{-1/2}x^{11/12+\varepsilon} + x^{1/2+\varepsilon})$$

as required. $\square$

In order to estimate the contribution of even d in Theorem 1 we need the following lemma.

Lemma 3. *Let q be an odd positive integer, let a be an integer with $(a(a+1), q) = 1$ and let $\varepsilon > 0$. Then*

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} r(n)r(n+1) = 8g(q)x + O(q^{-1/2}x^{11/12+\varepsilon} + x^{1/2+\varepsilon}),$$

with g as in Lemma 2 and the implied constant depending only on ε .

Proof. Exactly one of $n, n+1$ is odd, and for odd m one has $r(m) = 0$ unless $m \equiv 1 \pmod{4}$. If $n \equiv 2 \pmod{4}$ then $n+1 \equiv 3 \pmod{4}$, so $r(n+1) = 0$. Similarly, if $n \equiv 3 \pmod{4}$ then $r(n) = 0$. Hence only $n \equiv 0, 1 \pmod{4}$ contribute to the sum. We work these cases out separately.

First, let $n \equiv 0 \pmod{4}$ and denote its contribution by $S_1(x)$. As $n+1 \equiv 1 \pmod{4}$ we have

$$r(n+1) = 8 \sum_{\substack{k|n+1 \\ k < \sqrt{n+1}}} \chi_4(k) + 4\chi_4(\sqrt{n+1}),$$

and

$$\begin{aligned}
S_1(x) &= 8 \sum_{\substack{n \leq x \\ n \equiv a \pmod{q} \\ n \equiv 0 \pmod{4}}} r(n) \sum_{\substack{k|n+1 \\ k < \sqrt{n+1}}} \chi_4(k) + O(x^{1/2+\varepsilon}) \\
&= 8 \sum_{k < \sqrt{x+1}} \chi_4(k) \sum_{\substack{n \leq x \\ n \equiv 0 \pmod{4} \\ n \equiv a \pmod{q} \\ n \equiv -1 \pmod{k} \\ n > k^2-1}} r(n) + O(x^{1/2+\varepsilon}).
\end{aligned}$$

Here k is odd and $(k, q) = 1$ — otherwise the inner sum is empty, since a prime $p \mid (k, q)$ would divide $n + 1$ and q , hence $a + 1$, contrary to hypothesis. Writing $k \mid n + 1$ as $n \equiv -1 \pmod{k}$ and substituting $n = 4m$, so that $r(n) = r(m)$, the conditions on m become a single class $a' \pmod{qk}$ coprime to qk , and $(k^2 - 1)/4 < m \leq x/4$ (note that $(k^2 - 1)/4$ is an integer as k is odd). Since qk is odd, Lemma 1 applies

$$\begin{aligned}
\sum_{\substack{n \leq x \\ n \equiv 0 \pmod{4} \\ n \equiv a \pmod{q} \\ n \equiv -1 \pmod{k} \\ n > k^2-1}} r(n) &= \sum_{\substack{m \leq x/4 \\ m \equiv a' \pmod{qk} \\ m > (k^2-1)/4}} r(m) \\
&= \frac{\pi(x - k^2)}{4qk} h(q)h(k) + O((qk)^{-1/2}(x^{2/3+\varepsilon} + k^{4/3+\varepsilon})),
\end{aligned}$$

the $+1$ left over from $x - k^2 + 1$ being absorbed into the error term.

Summing the error over k we obtain

$$\begin{aligned}
\sum_{k < \sqrt{x+1}} (qk)^{-1/2}(x^{2/3+\varepsilon} + k^{4/3+\varepsilon}) &\ll q^{-1/2}x^{2/3+\varepsilon} \sum_{k \ll \sqrt{x}} k^{-1/2} \\
&\ll q^{-1/2}x^{11/12+\varepsilon}.
\end{aligned}$$

In the main term χ_4 vanishes at even k , thus

$$8 \sum_{\substack{k < \sqrt{x+1} \\ (k, q) = 1}} \chi_4(k) \frac{\pi(x - k^2)}{4qk} h(q)h(k) = \frac{2\pi h(q)}{q} \sum_{\substack{k < \sqrt{x+1} \\ (k, q) = 1}} \chi_4(k) h(k) \left(\frac{x}{k} - k \right).$$

As in the proof of Lemma 2 we have

$$\sum_{\substack{k \leq t \\ (k, q) = 1}} \chi_4(k) h(k) \ll \tau(q) \log t,$$

whence by partial summation

$$\sum_{\substack{k < \sqrt{x+1} \\ (k,q)=1}} \chi_4(k)h(k)k \ll \tau(q)x^{1/2} \log x, \quad \sum_{\substack{k \geq \sqrt{x+1} \\ (k,q)=1}} \frac{\chi_4(k)h(k)}{k} \ll \frac{\tau(q) \log x}{x^{1/2}}.$$

Since $h(q)\tau(q) \ll q^\varepsilon$, both contribute $O(x^{1/2+\varepsilon})$ after multiplication by $2\pi h(q)/q$ and $2\pi h(q)x/q$ respectively. On completing the sum to infinity we get

$$\sum_{\substack{k \geq 1 \\ (k,q)=1}} \frac{\chi_4(k)h(k)}{k} = \prod_{p \nmid q} \left(1 + \frac{\chi_4(p)}{p}\right) = \frac{2}{\pi} \prod_{p \mid q} \left(1 + \frac{\chi_4(p)}{p}\right)^{-1},$$

so that the main term equals

$$\frac{2\pi h(q)x}{q} \cdot \frac{2}{\pi} \prod_{p \mid q} \left(1 + \frac{\chi_4(p)}{p}\right)^{-1} = \frac{4x}{q} \prod_{p \mid q} \frac{p - \chi_4(p)}{p + \chi_4(p)} = 4g(q)x.$$

Therefore

$$S_1(x) = 4g(q)x + O(q^{-1/2}x^{11/12+\varepsilon} + x^{1/2+\varepsilon}).$$

Next, let $n \equiv 1 \pmod{4}$ and denote its contribution by $S_2(x)$. This time n itself is odd and $n \equiv 1 \pmod{4}$, so

$$r(n) = 8 \sum_{\substack{k \mid n \\ k < \sqrt{n}}} \chi_4(k) + 4\chi_4(\sqrt{n}),$$

therefore

$$S_2(x) = 8 \sum_{k < \sqrt{x}} \chi_4(k) \sum_{\substack{n \leq x \\ n \equiv 1 \pmod{4} \\ n \equiv a \pmod{q} \\ n \equiv 0 \pmod{k} \\ n > k^2}} r(n+1) + O(x^{1/2+\varepsilon}).$$

Again k is odd and $(k, q) = 1$, since a prime $p \mid (k, q)$ would divide n and q , hence a . Substituting $n+1 = 2m$, so that $r(n+1) = r(m)$, the condition $n \equiv 1 \pmod{4}$ makes m odd, the conditions on m become a single class $a'' \pmod{2qk}$ coprime to $2qk$, and $(k^2 + 1)/2 < m \leq (x+1)/2$. As $\chi_4(2) = 0$ we have $h(2qk) = h(q)h(k)$, and since $4 \nmid 2qk$

Lemma 1 gives

$$\begin{aligned}
\sum_{\substack{m \leq (x+1)/2 \\ m \equiv a'' \pmod{2qk} \\ m > (k^2+1)/2}} r(m) &= \frac{\pi}{2qk} \left(\frac{x+1}{2} - \frac{k^2+1}{2} \right) h(q)h(k) \\
&\quad + O((qk)^{-1/2}(x^{2/3+\varepsilon} + k^{4/3+\varepsilon})) \\
&= \frac{\pi(x-k^2)}{4qk} h(q)h(k) + O((qk)^{-1/2}(x^{2/3+\varepsilon} + k^{4/3+\varepsilon})).
\end{aligned}$$

Proceeding as in the case of $S_1(x)$ we conclude

$$S_2(x) = 4g(q)x + O(q^{-1/2}x^{11/12+\varepsilon} + x^{1/2+\varepsilon}).$$

Adding $S_1(x)$ and $S_2(x)$ completes the proof. $\square$

2.2. The linear sieve. Let $\mathcal{A} = (a_n)_{n \leq x}$ be a finite sequence of non-negative reals, let $\mathcal{P}$ be a set of primes, $P(z) = \prod_{p < z, p \in \mathcal{P}} p$, and

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{\substack{n \leq x \\ (n, P(z))=1}} a_n, \quad \mathcal{A}_d = \sum_{\substack{n \leq x \\ n \equiv 0 \pmod{d}}} a_n.$$

Suppose that there are $X > 0$, a multiplicative function g with $0 \leq g(p) < 1$ for $p \in \mathcal{P}$, and real numbers r_d such that

$$\mathcal{A}_d = g(d)X + r_d$$

for all square-free $d \mid P(z)$, and that g satisfies the linear sieve condition

$$\prod_{\substack{w \leq p < z \\ p \in \mathcal{P}}} (1 - g(p))^{-1} \leq \frac{\log z}{\log w} \left(1 + \frac{L}{\log w} \right)$$

for $2 \leq w \leq z$ and some constant $L \geq 1$. Put

$$V(z) = \prod_{\substack{p < z \\ p \in \mathcal{P}}} (1 - g(p)).$$

Lemma 4. *Let $D \geq z \geq 2$ and $s = \log D / \log z$. For $2 \leq s \leq 4$ define*

$$f(s) = \frac{2e^\gamma \log(s-1)}{s}.$$

Similarly, for $1 \leq s \leq 3$ set $F(s) = 2e^\gamma/s$. Under the above hypotheses we have

$$\begin{aligned} S(\mathcal{A}, \mathcal{P}, z) &\geq XV(z)(f(s) - B(\log D)^{-1/14}) - \sum_{\substack{d < D \\ d|P(z)}} |r_d|, \\ S(\mathcal{A}, \mathcal{P}, z) &\leq XV(z)(F(s) + B(\log D)^{-1/14}) + \sum_{\substack{d < D \\ d|P(z)}} |r_d|, \end{aligned}$$

where B depends only on L .

This is Theorem 8.4 of [4] (see also [5] and [6, Theorem 9.7]).

Finally we recall Mertens' formula in the weak form

$$\prod_{2 < p < z} \left(1 - \frac{1}{p}\right) \asymp \frac{1}{\log z}$$

which gives, for g as in Lemma 2,

$$V(z) = \prod_{2 < p < z} (1 - g(p)) \asymp \frac{1}{\log z},$$

since $\prod_{p>2} (1 - g(p))(1 - 1/p)^{-1}$ converges absolutely to a positive limit.

3. REDUCTION TO SUMS OF TWO SQUARES

Recall that $s_\Gamma(x)$ is the number of quadruples $(x_1, x_2, x_3, x_4) \in \mathbb{Z}^4$ subject to $n = x_1^2 + x_2^2 + x_3^2 + x_4^2 \leq x$, $x_1 x_4 - x_2 x_3 = 1$ and $\mu^2(n) = 1$.

Changing variables as in [2], we have

$$y_1 = x_1 + x_4, \quad y_2 = x_2 - x_3, \quad y_3 = x_1 - x_4, \quad y_4 = x_2 + x_3$$

or, equivalently,

$$x_1 = \frac{y_1 + y_3}{2}, \quad x_2 = \frac{y_2 + y_4}{2}, \quad x_3 = \frac{y_4 - y_2}{2}, \quad x_4 = \frac{y_1 - y_3}{2}.$$

We readily see that under this change of variables in s_Γ we have

$$y_1^2 + y_2^2 = n + 2, \quad (y_1^2 + y_2^2) - (y_3^2 + y_4^2) = 4, \quad \mu^2(n) = 1.$$

Note that if n is an odd integer, then $x_i \in \mathbb{Z}$ implies $y_i \in \mathbb{Z}$. Conversely, given y_i we have that y_1 has the opposite parity from y_2 , same holds for y_3 and y_4 . Then given y_i precisely one of the quadruples (y_1, y_2, y_3, y_4) and (y_1, y_2, y_4, y_3) gives rise to x_i . Thus the number of integer solutions in x 's is half of the number of integer solutions in y 's.

For even n the two sums $y_1^2 + y_2^2$ and $y_3^2 + y_4^2$ differ by 4, hence are congruent modulo 4, so either all four y_i are even or all four are odd — in both cases the x_i are integers and the correspondence is one-to-one.

4. SQUARE-FREE CASE: PROOF OF THEOREM 1

Let $r(n)$ denote the number of representations of n as the sum of two squares. The number of integer quadruples (y_1, y_2, y_3, y_4) subject to the above conditions equals $r(n+2)r(n-2)$, as the pairs (y_1, y_2) and (y_3, y_4) are chosen independently. By the above, the odd n contribute half of this number and the even n all of it, so that

$$(1) \quad s_\Gamma(x) = \frac{1}{2}s_\Gamma^{\text{odd}}(x) + s_\Gamma^{\text{even}}(x),$$

where

$$s_\Gamma^{\text{odd}}(x) = \sum_{\substack{n \leq x \\ n \text{ odd}}} \mu^2(n)r(n-2)r(n+2),$$

$$s_\Gamma^{\text{even}}(x) = \sum_{\substack{n \leq x \\ n \text{ even}}} \mu^2(n)r(n-2)r(n+2).$$

Notice that only $n \equiv 3 \pmod{4}$ contribute to $s_\Gamma^{\text{odd}}(x)$, since $r(n \pm 2) = 0$ for $n \equiv 1 \pmod{4}$. An even square-free n is of the form $n = 4m + 2$ with $2m + 1$ square-free, and then $n - 2 = 4m$, $n + 2 = 4(m + 1)$, so that $r(n - 2)r(n + 2) = r(m)r(m + 1)$ as $r(4k) = r(k)$, while $\mu^2(n) = \mu^2(2m + 1)$. Hence

$$s_\Gamma^{\text{even}}(x) = \sum_{m \leq (x-2)/4} \mu^2(2m + 1)r(m)r(m + 1).$$

The same decomposition holds for $s_\Gamma^{(k)}(x)$ with the additional condition $\omega(n) \leq k$ inserted throughout.

Below we estimate the odd and even contributions separately, see (2) and (3), which brings us to

$$s_\Gamma(x) = \left(\frac{1}{2}(8C) + 2C \right) x + O(x^{11/12+\varepsilon}) = 6Cx + O(x^{11/12+\varepsilon}).$$

4.1. Odd contribution. We begin with evaluating $s_\Gamma^{\text{odd}}(x)$. For $n \geq 3$ write $a_n = r(n - 2)r(n + 2)$, and put

$$s_\Gamma^{\text{odd}}(x) = \sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4}}} \mu^2(n)a_n, \quad s_{\Gamma,d}^{\text{odd}}(x) = \sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ n \equiv 0 \pmod{d}}} a_n.$$

Note that $a_n \geq 0$, and that $a_n \ll \tau(n + 2)\tau(n - 2) \ll x^\varepsilon$ for $n \leq x$. Since we sum over $n \equiv 3 \pmod{4}$, then $s_{\Gamma,d}^{\text{odd}}(x) = 0$ for even d , so we can freely impose the condition of d being odd.

By Lemma 2 we have

$$s_{\Gamma,d}^{\text{odd}}(x) = 8g(d)x + O(d^{-1/2}x^{11/12+\varepsilon} + x^{1/2+\varepsilon}).$$

Using $\mu^2(n) = \sum_{d^2|n} \mu(d)$ we obtain

$$s_{\Gamma}^{\text{odd}}(x) = \sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4}}} a_n \sum_{d^2|n} \mu(d) = \sum_{d \leq \sqrt{x}} \mu(d) s_{\Gamma,d^2}^{\text{odd}}(x).$$

We truncate the sum at $y = x^{1/4}$ and get

$$s_{\Gamma}^{\text{odd}}(x) = \sum_{d \leq y} \mu(d) s_{\Gamma,d^2}^{\text{odd}}(x) + O\left(\sum_{y < d \leq \sqrt{x}} s_{\Gamma,d^2}^{\text{odd}}(x)\right).$$

Bounding the summand in the error term trivially via

$$s_{\Gamma,d^2}^{\text{odd}}(x) \leq x^{\varepsilon} \sum_{\substack{n \leq x \\ n \equiv 0 \pmod{d^2}}} 1 \leq x^{\varepsilon} \left(\frac{x}{d^2} + 1\right)$$

we get

$$\sum_{y < d \leq \sqrt{x}} s_{\Gamma,d^2}^{\text{odd}}(x) \ll x^{\varepsilon} \sum_{y < d \leq \sqrt{x}} \left(\frac{x}{d^2} + 1\right) \ll x^{\varepsilon} \left(\frac{x}{y} + \sqrt{x}\right) \ll x^{3/4+\varepsilon}.$$

For the main sum we apply Lemma 2 with the odd modulus d^2 , which gives

$$\begin{aligned} \sum_{\substack{d \leq y \\ d \text{ odd}}} \mu(d) s_{\Gamma,d^2}^{\text{odd}}(x) &= 8x \sum_{\substack{d \leq y \\ d \text{ odd}}} \mu(d) g(d^2) + O(x^{11/12+\varepsilon} \log y + x^{1/2+\varepsilon} y) \\ &= 8x \sum_{\substack{d \leq y \\ d \text{ odd}}} \mu(d) g(d^2) + O(x^{11/12+\varepsilon}). \end{aligned}$$

Since $g(p^2) \leq 2/p^2$ we have

$$g(d^2) \leq \frac{2^{\omega(d)}}{d^2} \leq \frac{\tau(d)}{d^2} \ll \frac{1}{d^{2-\varepsilon}},$$

so the series $\sum_{d \text{ odd}} \mu(d) g(d^2)$ converges absolutely. On completing the sum to infinity we get

$$\sum_{\substack{d \geq 1 \\ d \text{ odd}}} \mu(d) g(d^2) = \prod_{p>2} (1 - g(p^2)) = \prod_{p>2} \left(1 - \frac{p - \chi_4(p)}{p^2(p + \chi_4(p))}\right) = C,$$

while the tail bounds as

$$\sum_{\substack{d > y \\ d \text{ odd}}} \mu(d) g(d^2) \ll \sum_{d > y} \frac{1}{d^{2-\varepsilon}} \ll \frac{1}{y^{1-\varepsilon}} \ll x^{-1/4+\varepsilon}.$$

Therefore we can conclude

$$(2) \quad s_{\Gamma}^{\text{odd}}(x) = 8Cx + O(x^{11/12+\varepsilon}).$$

4.2. Even contribution. We evaluate $s_{\Gamma}^{\text{even}}(x)$ via Lemma 3. Our goal is to show that for every $\varepsilon > 0$ we have $s_{\Gamma}^{\text{even}}(x) = 2Cx + O(x^{11/12+\varepsilon})$, with C as in Theorem 1. First, we have

$$s_{\Gamma}^{\text{even}}(x) = \sum_{n \leq X} \mu^2(2n+1) r(n) r(n+1)$$

with $X = (x-2)/4$. Expanding $\mu^2(2n+1) = \sum_{d^2 | 2n+1} \mu(d)$, with d odd, the condition $d^2 | 2n+1$ is equivalent to $n \equiv b \pmod{d^2}$ with $b = (d^2-1)/2$, and this class satisfies $2b \equiv -1$ and $2(b+1) \equiv 1 \pmod{d^2}$, so that $(b(b+1), d^2) = 1$. Since $d^2 | 2n+1 \leq 2X+1$, we obtain

$$s_{\Gamma}^{\text{even}}(x) = \sum_{d \leq \sqrt{2X+1}} \mu(d) \sum_{\substack{n \leq X \\ n \equiv b \pmod{d^2}}} r(n) r(n+1).$$

As in the case of the odd contribution take $y = x^{1/4}$. Bounding the inner sum trivially via

$$\sum_{\substack{n \leq X \\ n \equiv b \pmod{d^2}}} r(n) r(n+1) \leq x^{\varepsilon} \sum_{\substack{n \leq X \\ n \equiv b \pmod{d^2}}} 1 \leq x^{\varepsilon} \left(\frac{X}{d^2} + 1 \right),$$

we bound the tail by

$$\begin{aligned} \sum_{y < d \leq \sqrt{2X+1}} \sum_{\substack{n \leq X \\ n \equiv b \pmod{d^2}}} r(n) r(n+1) &\ll x^{\varepsilon} \sum_{y < d \leq \sqrt{2X+1}} \left(\frac{X}{d^2} + 1 \right) \\ &\ll x^{\varepsilon} \left(\frac{X}{y} + \sqrt{x} \right) \ll x^{3/4+\varepsilon}. \end{aligned}$$

Since d is odd, so is d^2 , and the implied constant in Lemma 3 depends only on ε , so the estimate it provides is uniform in d , the class b varying with d . As $X \ll x$, Lemma 3 with $q = d^2$ gives

$$\begin{aligned} (3) \quad s_{\Gamma}^{\text{even}}(x) &= 8X \sum_{\substack{d \leq y \\ d \text{ odd}}} \mu(d) g(d^2) + O(X^{11/12+\varepsilon} + yX^{1/2+\varepsilon} + x^{3/4+\varepsilon}) \\ &= 8CX + O(x^{11/12+\varepsilon}) = 2Cx + O(x^{11/12+\varepsilon}), \end{aligned}$$

since $\sum_{d \text{ odd}} \mu(d) g(d^2) = \prod_{p>2} (1 - g(p^2)) = C$ and the tail $d > y$ contributes $\ll xy^{-1+\varepsilon}$.

5. ALMOST PRIMES: PROOF OF THEOREM 3

We apply the linear sieve, i.e. Lemma 4 to the sequence

$$\mathcal{A} = (a_n)_{n \leq x}, \quad a_n = \begin{cases} r(n-2)r(n+2), & n \equiv 3 \pmod{4}, \\ 0, & \text{otherwise,} \end{cases}$$

with $\mathcal{P}$ the set of odd primes. The hypotheses of Section 2.2 hold with

$$X = 8x, \quad g(d) = \frac{1}{d} \prod_{p|d} \frac{p - \chi_4(p)}{p + \chi_4(p)}$$

and $r_d = \mathcal{A}_d - 8g(d)x$ for odd square-free d .

Note that $0 \leq g(p) < 1$, and the linear sieve condition is satisfied with some absolute L . By definition of g and Mertens' theorem we have

$$\log((1-g(p))^{-1}) = \frac{1}{p} + O(p^{-2}), \quad \sum_{w \leq p < z} \frac{1}{p} = \log\left(\frac{\log z}{\log w}\right) + O((\log w)^{-1}).$$

Let $0 < \theta \leq 1$ be such that $R(\theta)$ holds. By Lemma 2 we can always take $\theta = 1/6$, since for $D \leq x^{1/6-4\varepsilon}$

$$(4) \quad \sum_{\substack{d \leq D \\ d \text{ odd square-free}}} |r_d| \ll \sum_{d \leq D} (d^{-1/2} x^{11/12+\varepsilon} + x^{1/2+\varepsilon}) \\ \ll D^{1/2} x^{11/12+\varepsilon} + D x^{1/2+\varepsilon} \ll x^{1-\varepsilon}.$$

Note that using only the weaker error term $O(x^{11/12+\varepsilon})$, i.e. the one of [2, Lemma 4.2] one would get $\theta = 1/12$ here.

Take $N = \lfloor 2/\theta \rfloor$. Since $2/\theta < N+1$, we can fix $\varepsilon > 0$ and $0 < \eta \leq 2$ so that

$$\frac{2+\eta}{\theta-4\varepsilon} < N+1.$$

For $\theta = 1/6$ and $N = 12$ one may take $\varepsilon = 10^{-3}$ and $\eta = 10^{-2}$. Now set

$$D = x^{\theta-4\varepsilon}, \quad s = 2 + \eta, \quad z = D^{1/s}.$$

By $R(\theta)$ with $A = 2$, or by (4) when $\theta = 1/6$, we have

$$\sum_{\substack{d \leq D \\ d|P(z)}} |r_d| \ll \frac{x}{\log^2 x}.$$

Hence Lemma 4 together with $V(z) \asymp 1/\log z$ gives

$$S(\mathcal{A}, \mathcal{P}, z) \geq 8xV(z) \left(f(s) - B(\log D)^{-1/14} \right) - O(x(\log x)^{-2}) \gg \frac{x}{\log x}$$

for all sufficiently large x (here we used that $f(s) = 2e^\gamma \log(1+\eta)/(2+\eta) > 0$).

Every n with $a_n \neq 0$ and $(n, P(z)) = 1$ is odd and has all its prime factors at least z . Since $n \leq x$, the number $\Omega(n)$ of prime factors of n counted with multiplicity satisfies

$$\Omega(n) \leq \frac{\log x}{\log z} = \frac{s}{\theta - 4\varepsilon} = \frac{2 + \eta}{\theta - 4\varepsilon} < N + 1,$$

by the choice of ε and η , so that $\Omega(n) \leq N$.

It remains to discard the n that are not square-free. Such an n counted in $S(\mathcal{A}, \mathcal{P}, z)$ is divisible by p^2 for some prime p with $z \leq p \leq \sqrt{x}$, and by the trivial bound for r

$$\sum_{z \leq p \leq \sqrt{x}} \sum_{\substack{n \leq x \\ p^2 | n}} a_n \ll x^\varepsilon \sum_{z \leq p \leq \sqrt{x}} \left(\frac{x}{p^2} + 1 \right) \ll x^\varepsilon \left(\frac{x}{z} + \sqrt{x} \right) = o\left(\frac{x}{\log x} \right),$$

because z is a fixed positive power of x . Hence

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ \mu^2(n)=1 \\ \omega(n) \leq N}} r(n-2)r(n+2) \geq S(\mathcal{A}, \mathcal{P}, z) - o\left(\frac{x}{\log x} \right) \gg \frac{x}{\log x},$$

where we used that $\omega(n) = \Omega(n)$ for square-free n . By (1), with the condition $\omega(n) \leq N$ inserted, the left-hand side is at most $2s_\Gamma^{(N)}(x)$, and the lower bound $s_\Gamma^{(N)}(x) \gg x/\log x$ follows for large x (for x bounded it is trivial since $s_\Gamma^{(N)}(x) \geq s_\Gamma^{(N)}(3) > 0$). This finishes the proof of Theorem 3. Taking $\theta = 1/6$ and $N = 12$ gives $s_\Gamma^{(12)}(x) \gg x/\log x$ unconditionally.

Remark 1. The exponent $1/6$ in the unconditional level of distribution comes from the individual error term $O(d^{-1/2}x^{11/12+\varepsilon})$ in Lemma 2, which in turn comes from the summation of terms coming from Lemma 1 over the moduli $4dk$ with $k < \sqrt{x}$. The value 7 is what the level $1/6$ allows. By Theorem 3 the bound $\omega(n) \leq k$ requires the stronger condition

$$\theta > \left(k + 1 - \frac{\log 4}{\log 3} \right)^{-1},$$

which $\theta = 1/6$ satisfies for $k = 7$ and fails for $k = 6$. Thus the number 7 is not improved by any rearrangement of the sieve, only by a larger level of distribution.

6. THE WEIGHTED SIEVE: PROOF OF THEOREM 2

We keep the notation of Section 5, see [3, Chapter 5] for the general theory of the weighted sieve. Set $\mathcal{A} = (a_n)$ with $a_n = r(n-2)r(n+2)$ for $n \leq x$, $n \equiv 3 \pmod{4}$ and $a_n = 0$ otherwise, $\mathcal{P}$ the odd primes, $X = 8x$, g as in Lemma 2, $r_d = \mathcal{A}_d - 8g(d)x$, and

$$V(z) = \prod_{2 < p < z} (1 - g(p)) \asymp \frac{1}{\log z}.$$

Fix $\varepsilon > 0$ and put

$$\alpha = 1/6 - 4\varepsilon, \quad D = x^\alpha, \quad z = D^{1/4} = x^{\alpha/4}, \quad y = x^{3/20}, \quad \lambda = 4/3.$$

Taking ε small enough so that $\alpha - 3/20 \geq 1/100$ (i.e. $\varepsilon \leq 1/600$) we have $z < y < D$ and $D/y = x^{\alpha-3/20} \geq x^{1/100}$. By (4), the remainders satisfy $\sum_{m \leq D} |r_m| \ll x^{1-\varepsilon}$, where the sum is taken over odd square-free m .

Consider the weighted sum

$$W(x) = \sum_{\substack{n \leq x \\ (n, P(z))=1}} a_n b_n, \quad \text{where } b_n = 1 - \frac{1}{\lambda} \sum_{\substack{p|n \\ z \leq p < y}} \left(1 - \frac{\log p}{\log y}\right).$$

Let us begin with interpreting what $b_n > 0$ implies for us. Let n be square-free with $(n, P(z)) = 1$ and $b_n > 0$. All prime factors of n are at least z . For $p \geq y$ the quantity $1 - \log p / \log y$ is at most 0, hence

$$\begin{aligned} \sum_{\substack{p|n \\ p < y}} \left(1 - \frac{\log p}{\log y}\right) &\geq \sum_{p|n} \left(1 - \frac{\log p}{\log y}\right) = \omega(n) - \frac{\log n}{\log y} \\ &\geq \omega(n) - \frac{\log x}{\log y} = \omega(n) - \frac{20}{3}. \end{aligned}$$

Therefore $b_n > 0$ forces $\omega(n) - 20/3 < \lambda = 4/3$, that is $\omega(n) < 8$. Since $b_n \leq 1$ for every n , it follows that

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ \mu^2(n)=1 \\ \omega(n) \leq 7}} r(n-2)r(n+2) \geq \sum_{\substack{n \leq x \\ (n, P(z))=1 \\ \mu^2(n)=1}} a_n b_n \geq W(x) - \sum_{\substack{n \leq x \\ (n, P(z))=1 \\ \mu^2(n)=0}} a_n |b_n|.$$

For $n \leq x$ with all prime factors at least z we have

$$|b_n| \leq 1 + \frac{\omega(n)}{\lambda} \leq 1 + \frac{\log x}{\lambda \log z} \ll 1,$$

and such an n that is not square-free is divisible by p^2 for some prime p with $z \leq p \leq \sqrt{x}$. Hence, by the trivial bound for r , the last sum

above is $\ll x^\varepsilon(x/z + \sqrt{x}) = o(x/\log x)$. It therefore remains to prove that $W(x) \gg x/\log x$.

We are now ready to apply linear sieve, i.e. Lemma 4 and bound $W(x)$. Interchanging the order of summation we get

$$W(x) = S(\mathcal{A}, \mathcal{P}, z) - \frac{1}{\lambda} \sum_{z \leq p < y} \left(1 - \frac{\log p}{\log y}\right) S(\mathcal{A}_p, \mathcal{P}, z),$$

where $\mathcal{A}_p = (a_n)_{p|n}$ and $S(\mathcal{A}_p, \mathcal{P}, z) = \sum_{p|n, (n, P(z))=1} a_n$. Note that $p \geq z$, so the condition $(n, P(z)) = 1$ does not exclude $p | n$.

For the first term we apply Lemma 4 with level $D = z^4$, so that $s = 4$ and $f(4) = \frac{e^\gamma}{2} \log 3$, which gives

$$S(\mathcal{A}, \mathcal{P}, z) \geq XV(z) \left(\frac{e^\gamma}{2} \log 3 - B(\log D)^{-1/14} \right) - \sum_{\substack{d < D \\ d|P(z)}} |r_d|.$$

For $z \leq p < y$ and odd square-free d coprime to p we have

$$\sum_{\substack{n \leq x \\ n \equiv 0 \pmod{pd}}} a_n = 8g(pd)x + r_{pd} = 8g(p)xg(d) + r_{pd},$$

so the sequence $\mathcal{A}_p$ satisfies the hypotheses of Section 2.2 with $X_p = 8g(p)x$, the same density g , and remainders r_{pd} . Its level of distribution is $D_p = D/p$, and

$$s_p = \frac{\log D_p}{\log z} = 4 \left(1 - \frac{\log p}{\log D}\right)$$

ranges from $4(1 - 3/(20\alpha)) < 1$ to 3. Further, Lemma 4 applies when $s_p \geq 1$, so we first work out the case $s_p < 1$. We have

$$S(\mathcal{A}_p, \mathcal{P}, z) \leq S(\mathcal{A}_p, \mathcal{P}, D_p).$$

By the sieve condition we have

$$\frac{V(D_p)}{V(z)} = \prod_{D_p \leq p < z} (1 - g(p))^{-1} \leq \frac{\log z}{\log D_p} \left(1 + \frac{L}{\log D_p}\right) = s_p^{-1}(1 + o(1)),$$

where the error is uniform as $D_p \geq D/y \geq x^{1/100}$, therefore Lemma 4 applies.

Now we have $s_p \geq 1$ and therefore can apply Lemma 4, the latter applies because $D_p \geq D/y \geq x^{1/100}$. In both cases

$$S(\mathcal{A}_p, \mathcal{P}, z) \leq 8g(p)xV(z) \frac{2e^\gamma}{s_p} (1 + o(1)) + \sum_{\substack{d < D/p \\ d \text{ odd square-free}}} |r_{pd}|,$$

where the term $o(1)$, which comes from $B(\log(D/p))^{-1/14}$ and from $L/\log D_p$, is uniform in p because $D_p \geq x^{1/100}$. Every odd square-free $m < D$ occurs as $m = pd$ for at most $\omega(m) \ll \log D$ primes p , so the total contribution of all the remainder terms above to $W(x)$ is $\ll x^{1-\varepsilon} \log x = o(x/\log x)$.

We proceed with summing over p . Write $t = \log p / \log x$, so that the condition $z \leq p < y$ becomes $\alpha/4 \leq t < 3/20$, while $1/s_p = \alpha/(4(\alpha-t))$ and $1 - \log p / \log y = 1 - 20t/3$. Let ϕ be any function that is continuously differentiable on $[\alpha/4, 3/20]$. Since $g(p) = 1/p + O(p^{-2})$ and $\sum_{p \leq v} 1/p = \log \log v + c_0 + O(1/\log v)$, partial summation gives

$$\sum_{z \leq p < y} g(p) \phi\left(\frac{\log p}{\log x}\right) = \int_{\alpha/4}^{3/20} \phi(t) \frac{dt}{t} + O\left(\frac{1}{\log x}\right),$$

the implied constant depending on ϕ . Take

$$\phi(t) = \left(1 - \frac{20}{3}t\right) \cdot 2e^\gamma \frac{\alpha}{4(\alpha-t)},$$

which is smooth on the interval $[\alpha/4, 3/20]$ since $\alpha - t \geq \alpha - 3/20 > 0$. Then we have

$$\frac{1}{\lambda} \sum_{z \leq p < y} \left(1 - \frac{\log p}{\log y}\right) S(\mathcal{A}_p, \mathcal{P}, z) \leq XV(z) \left(\frac{1}{\lambda} J(\alpha) + o(1)\right) + o\left(\frac{x}{\log x}\right),$$

where

$$J(\alpha) = \frac{e^\gamma \alpha}{2} \int_{\alpha/4}^{3/20} \frac{1 - 20t/3}{t(\alpha-t)} dt.$$

Setting $\eta = \log y / \log x = 3/20$ inside the integral we have

$$\frac{\alpha(1-t/\eta)}{t(\alpha-t)} = \frac{1}{t} - \frac{\alpha-\eta}{\eta} \cdot \frac{1}{\alpha-t},$$

hence

$$J(\alpha) = \frac{e^\gamma}{2} \left(\log \frac{4\eta}{\alpha} - \frac{\alpha-\eta}{\eta} \log \frac{3\alpha}{4(\alpha-\eta)} \right).$$

Both terms depend continuously on α near $1/6$, and for $\alpha = 1/6$, that is for $\varepsilon = 0$, we have $4\eta/\alpha = 18/5$, $(\alpha-\eta)/\eta = 1/9$ and $3\alpha/(4(\alpha-\eta)) = 15/2$, whence

$$J(1/6) = \frac{e^\gamma}{2} \left(\log \frac{18}{5} - \frac{1}{9} \log \frac{15}{2} \right).$$

Combining the splitting of $W(x)$ with the two bounds just obtained, we get

$$W(x) \geq XV(z) \left(\frac{e^\gamma}{2} \log 3 - \frac{3}{4} J(\alpha) - o(1) \right) - o\left(\frac{x}{\log x}\right).$$

At $\alpha = 1/6$ the constant becomes

$$\frac{e^\gamma}{2} \left(\log 3 - \frac{3}{4} \log \frac{18}{5} + \frac{1}{12} \log \frac{15}{2} \right) = \frac{e^\gamma}{24} \log \frac{3^{12} \cdot (15/2)}{(18/5)^9} > 0.$$

By continuity in α we may fix $\varepsilon > 0$ so small that

$$\frac{e^\gamma}{2} \log 3 - \frac{3}{4} J(\alpha) > \frac{5e^\gamma}{48} \log \frac{25}{12},$$

and then, for large x ,

$$W(x) \geq \frac{5e^\gamma}{48} \log \frac{25}{12} \cdot XV(z) \gg \frac{x}{\log x},$$

since $V(z) \asymp 1/\log z$. This gives

$$\sum_{\substack{n \leq x \\ n \equiv 3 \pmod{4} \\ \mu^2(n)=1 \\ \omega(n) \leq 7}} r(n-2)r(n+2) \gg \frac{x}{\log x},$$

and by (1) the left-hand side is at most $2s_\Gamma^{(7)}(x)$, which proves Theorem 2 for large x . For bounded x the desired conclusion follows trivially from $s_\Gamma^{(N)}(x) \geq s_\Gamma^{(N)}(3) > 0$.

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